J Vijayavallabh

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Summary

Final year **B.Tech (Biological Engineering), IIT Madras**; focused on **AI for biotech** and **multimodal/agentic systems**.

Education

2027 **B.Tech Biological Engineering**, **IIT Madras** (CGPA: 8.64/10)
2023 Class XII (CBSE), **Sri Chaitanya Techno School** (95.4%)
2021 Class X (CBSE), **Sri Chaitanya Techno School** (99.4%)

Publications and Patent

- **The Regularizer That Switched Off the Experiment: Why Our Protein-Interaction Null Was About the Encoder, Not the Prior**. **NeurIPS** 2026 Workshop on Failure Modes of AI in Biology (**ICBINB-BIO**), Sydney. Poster.
- **Forced Choice Is Not a Tolerance: What Multiple-Choice Grading Accepts on BixBench's Numeric Questions**. **NeurIPS** 2026 Workshop on Agentic AI for Biological Discovery (**AgenticLS**) Sydney. Poster.
- **An Empirical Audit of k-NAF Budget Accounting for Anchored Decoding** (arXiv:2605.28001 preprint).
- **Enhancing Financial RAG with Agentic AI and Multi-HyDE** (Proceedings of **EMNLP** 2025 FinNLP) [Inter-IIT].
- **Sparse Hyperbolic Convolutional Networks with Enhanced Object Localization via GradCAM** (Proceedings of International Conference on Computer Vision **ICCV** Beyond Euclidean Workshop **BEW** 2025) [SENAI IITM].
- **Patent (granted):** SYSTEMS AND METHODS FOR AGENTIC BASED APPROACH IN RETRIEVAL AUGMENTED GENERATION

Professional Experience

AI Research and Development Intern, Photios Apr 2025 - May 2025

- Fine-tuned state-of-the-art Automatic Speech Recognition (**ASR**) models using Deep Learning techniques for transcribing sales interactions in retail environments, by extensive literature review and comparative analysis of ASR architectures.
- Architected an automated data cleaning pipeline with the **FastText** model to filter Sanskrit words from Hindi datasets, then fine-tuned custom **N-Gram language model** integrated with **AI4Bharat Conformer**, reducing Word Error Rate (**WER**).

Positions of Responsibility

Marketing and Operations Coordinator, Finance Club, IIT Madras Jul 2024 to Mar 2025

- Planned and delivered the club's flagship **36th Wall Street** finance fest (7 to 9 Feb 2025), coordinating **8+ sessions** across keynotes, workshops, and competitions; handled speaker outreach, hospitality, scheduling, and on-ground logistics across CLT, DoMS, and CRC, including sessions involving **CFA Institute**, **IIQF**, **CAMS**, and **GetBit**, as well as IIPA-led sessions.
- Invited and hosted chief guests **Arun Kohli** (MD & Country Head, India, Morgan Stanley) and **Shankar Sharma** (Co-founder, First Global; Founder, GQuant); aligned session themes, moderated fireside chats and Q&A, and managed the guest experience.
- Ran marketing and operations for club programming. Publicized events through institute mailing lists, Unstop, WhatsApp communities, LinkedIn, Instagram, and Facebook; launched **6+ competitions** with **StockGro**, **Finlatics**, and **QuantInsti**; ran campus editions of **IMC Prosperity** and **WorldQuant BRAIN IQC 2025**; co-organized **Innovision Hackathon 2024 to 2025** with **Indian Bank**, CIFIL, and DFS; managed **5** volunteers and a **Rs. 4.8 lakh** budget, and secured sponsorships.

Research Experience

Adversarial Auditing of K-NAF Privacy [Privacy in AI Course Project] Jan - May 2026

- Built an adversarial auditing pipeline for privacy-aware text generation to test K-NAF decoding guarantees under targeted prompt attacks across optimization, validation, stress tests, and held-out evaluation over multiple global K budgets and model variants.
- Measured certified privacy risk via U_{EBB}, effective budget, and ρ; both settings stayed certified in aggregate but $k = 3$ showed stronger held-out transfer, sharper attack separation curves, and more reliable audit validity than $k = 5$ overall.

Research Intern, Kansas State University [Paper under review] May - Jul 2025

- Developed **Position-Aware Inductive Graph Transformer (PAIGT)** for scRNA-seq (Prof. Bala Natarajan), GraphSAGE with degree/shortest-path/path-attention encoders and 6 biological-prior modules (18 CLI controls) over 4 injection points.
- Built leak-free pipeline (split-before-HVG; train-only HVG/kNN/prior std.) with **scTEP** ensemble inference; ran a **12,576-config** $\times$ **5-seed** sweep (23 arms, 7 metrics, 88,032) on 4 GPUs across 18 cohorts, boosting acc. **+4.3 pp** and ARI/NMI **+0.59/+0.48**.

Student Researcher, SENAI IITM Dec 2024 - May 2025

- Worked on **geometric representation learning** (Prof. Raghunathan Rengaswamy): Hyperbolic Grad-CAM, activation sparsity in Hyperbolic/hybrid CNNs for interpretability in non-Euclidean spaces, revealing hierarchical/part-whole structure in images.
- Delivered **technical talks**: geometric deep learning, disentanglement via topology, topology of DNNs, and PCA in ML research.

Student Researcher, RBCDSAI Sep 2024 - Dec 2024

- Modelled **reaction-protein relationships** for reaction prediction (Prof. Karthik Raman) by building a **seq2seq** pipeline with **ESM**-encoded proteins and a **BART**-style cross-attention decoder to generate all SMILES sequences for a given reaction sequence.
- Optimized the model capabilities through tokenization and data partitioning experiments to maximize chemically valid SMILES.

Personal Projects

Anthropic $\times$ Adaptyv Protein Design Challenges Oct 2026
EGFR | TNF-α | GLP-1R

- Built pipelines combining RFdiffusion, ProteinMPNN, and structure prediction to generate 58 protein designs targeting EGFR, TNF-α, and GLP-1R; automated checks for structural confidence, novelty, and predicted binding to human and mouse targets.

- Tested pH-dependent binding via MD and free-energy: 12/20 EGFR designs passed 4 checks; 216 TNF simulations yielded five candidates binding at blood pH and releasing in acidic conditions (human/mouse). Robust controls flagged a GLP-1R score.

Forced Choice Is Not a Tolerance (NeurIPS 2026 AgenticLS Workshop) Sep 2026

- Performed a reproducible audit of **BixBench** numeric evaluation by regrading **5k+** public runs and new agent runs against a **5% tolerance** baseline; showed forced-choice MCQ grading inflates scores by accepting 11–22 points worth of incorrect answers.
- Built analyses to attribute error sources (nearest-option bias, **key-rank leakage**, refusal/parse failures) and stress-tested **counterfactual option rewrites**; delivered benchmark-design recommendations and plots for more reliable agent comparisons.

The Regularizer That Switched Off the Experiment (NeurIPS 2026 ICBINB-BIO Workshop) July 2026

- Shipped **EG-IPG** (evidence-gated relational GNN) on an **8M-edge** PPI graph (**ESM-2**+**PINNACLE**) for CRISPRi Perturb-seq response prediction, with reproducible runs and clear ablations throughout end-to-end; **NeurIPS 2026** workshop poster.
- Productionized a **multi-GPU** training/eval stack (sweeps, run registry, tests) and led leakage-safe **OOD** benchmarking across **8** Perturb-seq datasets; debugged a critical gating regularizer issue and isolated the **encoder** bottleneck for robust model selection.

Inter IIT Tech Meet 14.0 (ISRO GeoNLI) Nov - Dec 2025

- Led a 6-member team representing IIT Madras in a **Mid Prep PS** over a span of 15 days to build ***GeoNLI***, an offline/on-prem remote-sensing Vision Language Model [**VLM**] system for **captioning**, semantic/binary/numeric **VQA**, and visual **grounding**.
- Curated **multi-resolution RS** datasets (VRSBench, RSVLM-QA, XLRS) and synthesized adversarial + counting/area VQA.
- Finetuned **Ovis2.5-9B** using **LoRA** on captioning and VQA, and implemented a **hybrid grounding** stack [also in similar fashion for VQA] (Ovis intent routing → **SAM-3** segmentation → OpenCV mask-metadata → OBB + tool-based geometry).

Inter IIT Tech Meet 13.0 (Pathway) Oct - Dec 2024

- Represented IIT Madras in a **High Prep PS** to build a novel dynamic agentic Retrieval Augmented Generation [**RAG**] pipeline for financial documents like **SEC-10K** filings with **adaptive, secure, explainable** workflows using Pathway's vector store.
- Engineered **Multi-HyDE** hybrid retrieval mechanism combining multi-query and hypothetical document embeddings, robust **PDF parsing**, Llama Guard-based **safety guardrails**, and **explainable** AI source attribution; deployed on **Azure** (Docker + FastAPI) and evaluated on **GPT-4o/Gemini-1.5** using semantic similarity, recall, factual correctness, faithfulness, and ROUGE.

Scholastic Achievements

- **Top 1%** in JEE Main and JEE Advanced 2023
- **Reviewer:** UniReps (**NeurIPS** 2025 Workshop)
- **NeurIPS 2026:** Received **financial aid**
- **Google DeepMind APAC Research Symposium 2026:** Selected to join a cohort of **˜200** researchers across undergraduate, master's, and final-year PhD levels for a two-day AI research symposium.
- **YC Startup School India (Bengaluru)** – Received **YC AI Stack** benefits (*$25k+* in AI tools + cloud credits)
- **FAST Grants (Biswas Family Foundation)** – **$25,000** Fast Grant (Cycle 1; decision *Jul 2026*, funded *˜Aug 2026*)
  PhageHost-Bench: open benchmark for AI-driven phage–host prediction (AI ∩ health; 12-month project)
- **Top 6/23** – Inter IIT Tech Meet 13.0 (High Prep)
  Dynamic Agentic RAG for SEC-10K filings
- **Top 8/23** – Inter IIT Tech Meet 14.0 (Mid Prep)
  ISRO GeoNLI
- **ABRSM Grade 8 Piano Practical** – Merit (Nov 2024)

Skills

Technical	**Python**, Matlab, **PyTorch**, TensorFlow, Pandas, NumPy, RDKit, DGL, NetworkX, **LangChain**, Unsloth, MNE-Python, Optuna, NLTK, Matplotlib, Git/GitHub, Docker, FastAPI, Bash, Azure, Pix2pixHD, LaTeX
Courses/Certs	Protein Structures and Functions, Computational Systems Biology, Bioinformatics, Privacy in AI, Computational Neuroscience, Microbiology and Biochemistry, Principles of Neuroscience, DSA for Biology, Molecular Biology and Genetic Engineering, Biological Engineering Lab, Advanced Topics in AI [Large Scale Cloud Computing and Federated Learning], Signals and Systems, Linear Algebra, Probability and Statistics.